Week .3

Course Name : Problem Solving and Testing

Course Code : 10211CS227

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1. Java Date and Time
2. Number of Days Between Two Dates
3. Day of the Year
4. Day of the Week
5. Java Priority Queue
6. Java Arraylist
7. Largest Number
8. Java Comparator
9. Sort the People

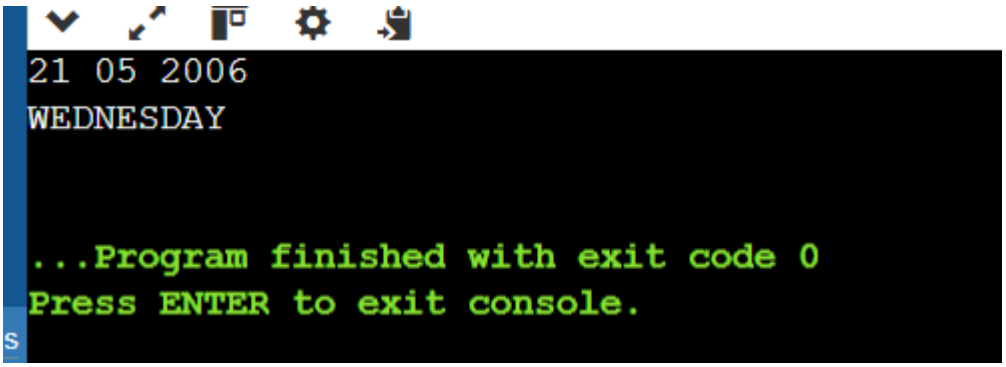
1. Java Date and Time

PROGRAM:

```
import java.util.*;

public class Main {

    public static String findDay(int month, int day, int year) {
        Calendar cal = Calendar.getInstance();

        cal.set(year, month - 1, day);

        String[] days = {
            "SUNDAY",
            "MONDAY",
            "TUESDAY",
            "WEDNESDAY",
            "THURSDAY",
            "FRIDAY",
            "SATURDAY"
        };

        return days[cal.get(Calendar.DAY_OF_WEEK) - 1];
    }

    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        int month = sc.nextInt();
        int day = sc.nextInt();
        int year = sc.nextInt();

        System.out.println(findDay(month, day, year));

        sc.close();
    }
}
```

OUTPUT:

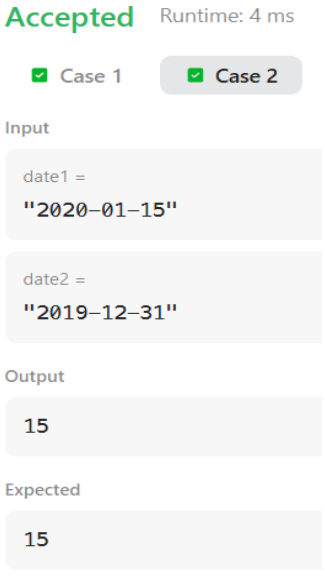


2. Number of Days Between Two Dates

PROGRAM:

```
import java.time.LocalDate;
import java.time.temporal.ChronoUnit;

class Solution {
    public int daysBetweenDates(String date1, String date2) {
        LocalDate d1 = LocalDate.parse(date1);
        LocalDate d2 = LocalDate.parse(date2);

        return (int) Math.abs(ChronoUnit.DAYS.between(d1, d2));
    }
}
```

OUTPUT:

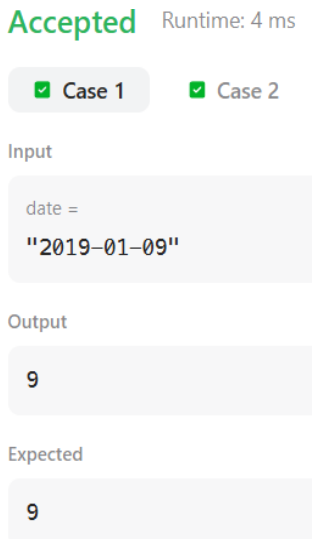


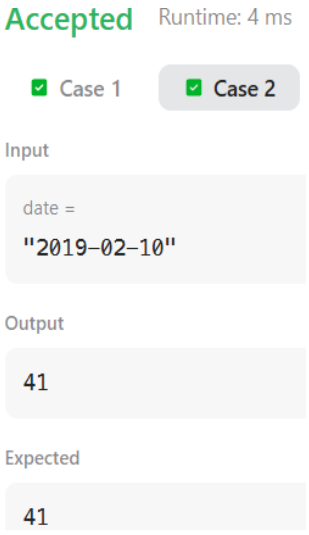
3. Day of the Year

PROGRAM:

```
import java.time.LocalDate;

class Solution {
    public int dayOfYear(String date) {
        LocalDate d = LocalDate.parse(date);
        return d.getDayOfYear();
    }
}
```

OUTPUT:



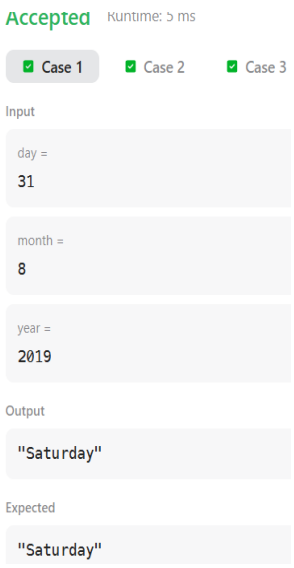


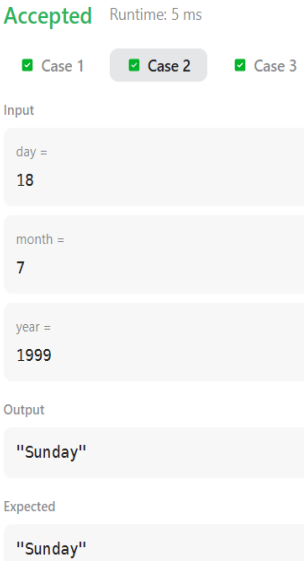
4. Day of the Week

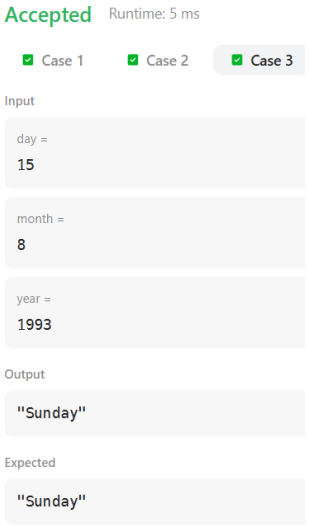
PROGRAM:

```
import java.time.LocalDate;

class Solution {
    public String dayOfTheWeek(int day, int month, int year) {
        LocalDate date = LocalDate.of(year, month, day);

        String dayName = date.getDayOfWeek().toString();

        return dayName.charAt(0) + dayName.substring(1).toLowerCase();
    }
}
```

OUTPUT:







5. Java Priority Queue

PROGRAM:

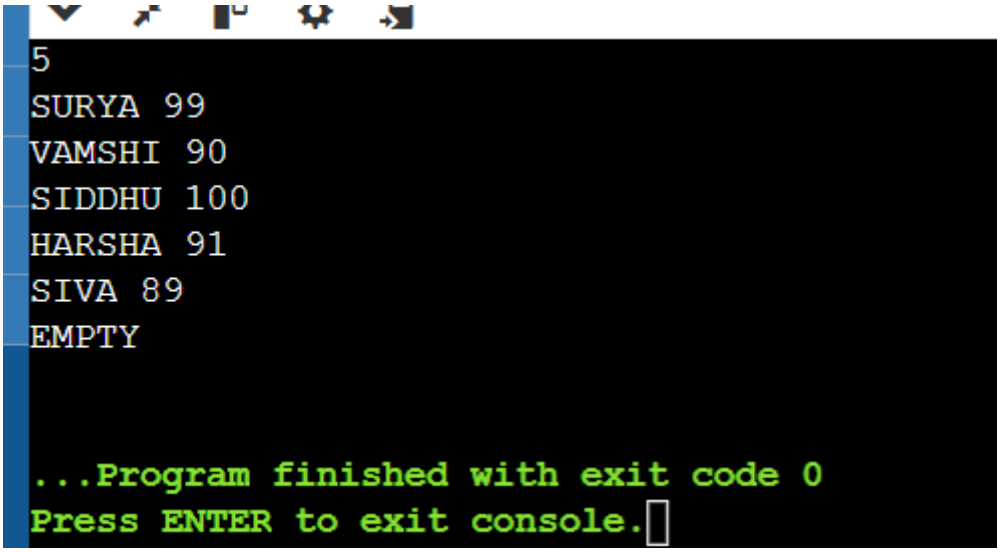
```
import java.util.*;

class Student {
    private int id;
    private String name;
    private double cgpa;

    public Student(int id, String name, double cgpa) {
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }

    public int getID() {
        return id;
    }

    public String getName() {
        return name;
    }

    public double getCGPA() {
        return cgpa;
    }
}
```

```
class Priorities {

    public List<Student> getStudents(List<String> events) {

        PriorityQueue<Student> pq = new PriorityQueue<>(
            (a, b) -> {
                if (a.getCGPA() != b.getCGPA()) {
                    return Double.compare(b.getCGPA(), a.getCGPA());
                }

                int nameCompare = a.getName().compareTo(b.getName());

                if (nameCompare != 0) {
                    return nameCompare;
                }

                return Integer.compare(a.getID(), b.getID());
            }
        );

        for (String event : events) {

            if (event.startsWith("ENTER")) {
                String[] parts = event.split(" ");

                String name = parts[1];
```

```
            double cgpa = Double.parseDouble(parts[2]);
            int id = Integer.parseInt(parts[3]);

            pq.add(new Student(id, name, cgpa));

        } else if (event.equals("SERVED")) {
            if (!pq.isEmpty()) {
                pq.poll();
            }
        }
    }

    List<Student> result = new ArrayList<>();

    while (!pq.isEmpty()) {
        result.add(pq.poll());
    }

    return result;
  }
}

public class Main {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
        sc.nextLine();

        List<String> events = new ArrayList<>();

        for (int i = 0; i < n; i++) {
            events.add(sc.nextLine());
        }

        Priorities priorities = new Priorities();
        List<Student> students = priorities.getStudents(events);

        if (students.isEmpty()) {
            System.out.println("EMPTY");
        } else {
            for (Student student : students) {
                System.out.println(student.getName());
            }
        }

        sc.close();
    }
}
```

OUTPUT:



6.Java Arraylist

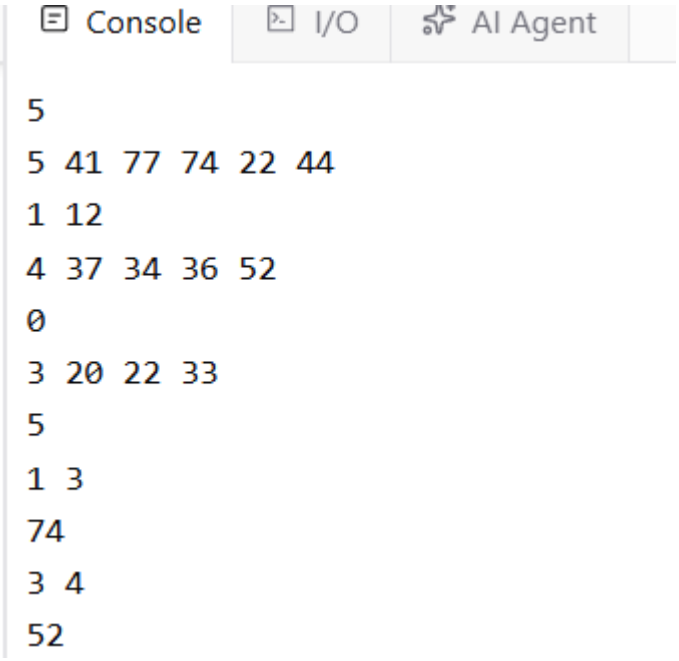
PROGRAM:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

ArrayList<ArrayList<Integer>> list = new ArrayList<>();

// Read the lines
for (int i = 0; i < n; i++) {
    int size = sc.nextInt();

    ArrayList<Integer> row = new ArrayList<>();

    for (int j = 0; j < size; j++) {
        row.add(sc.nextInt());
    }

    list.add(row);
}

// Number of queries
int q = sc.nextInt();

for (int i = 0; i < q; i++) {
    int x = sc.nextInt();
    int y = sc.nextInt();
```

```
            // x and y are 1-based
            if (x >= 1 && x <= list.size() &&
                y >= 1 && y <= list.get(x - 1).size()) {

                System.out.println(list.get(x - 1).get(y - 1));

            } else {
                System.out.println("ERROR!");
            }
        }

        sc.close();
    }
}
```

OUTPUT:



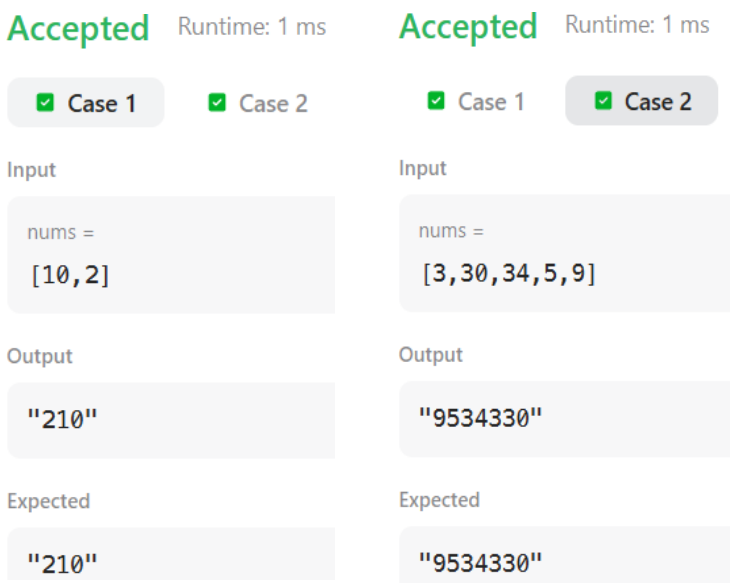
7. Largest Number

PROGRAM :

```
import java.util.*;

class Solution {
    public String largestNumber(int[] nums) {

      String[] arr = new String[nums.length];

     for (int i = 0; i < nums.length; i++) {
         arr[i] = String.valueOf(nums[i]);
     }

     Arrays.sort(arr, (a, b) -> (b + a).compareTo(a + b));

     if (arr[0].equals("0")) {
         return "0";
     }

     StringBuilder result = new StringBuilder();

     for (String s : arr) {
         result.append(s);
     }

     return result.toString();
```

```
    }
}
```

OUTPUT :



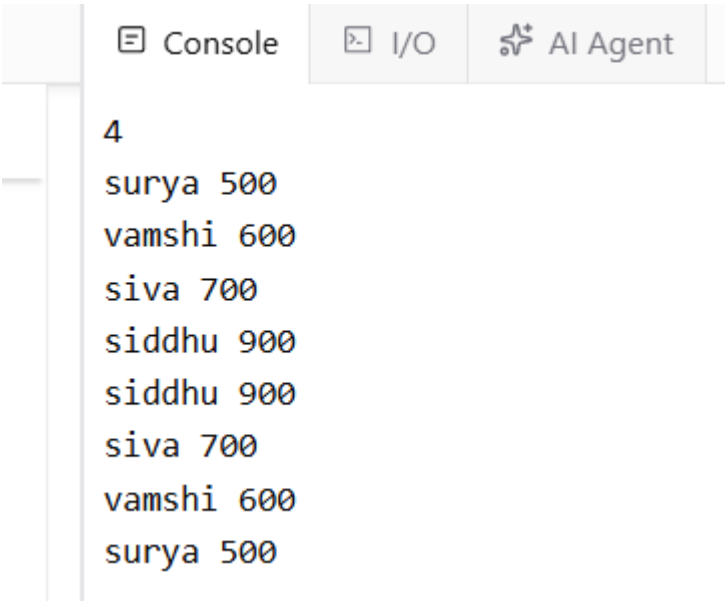
8.Java Comparator

PROGRAM :

```
import java.util.*;

class Player {
    String name;
    int score;

    Player(String name, int score) {
        this.name = name;
        this.score = score;
    }
}

class Checker implements Comparator<Player> {

    @Override
    public int compare(Player a, Player b) {

        // Sort by decreasing score
        if (a.score != b.score) {
            return b.score - a.score;
        }

        // If scores are same, sort alphabetically
        return a.name.compareTo(b.name);
    }
```

```
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        Player[] players = new Player[n];

        for (int i = 0; i < n; i++) {
            String name = sc.next();
            int score = sc.nextInt();

            players[i] = new Player(name, score);
        }

        Arrays.sort(players, new Checker());

        for (Player p : players) {
            System.out.println(p.name + " " + p.score);
        }

        sc.close();
    }
}
```

OUTPUT :



9. Sort the People

PROGRAM :

```
import java.util.*;
```

```
class Solution {
    public String[] sortPeople(String[] names, int[] heights) {

        Integer[] index = new Integer[names.length];

        for (int i = 0; i < names.length; i++) {
            index[i] = i;
        }

        Arrays.sort(index, (a, b) -> heights[b] - heights[a]);

        String[] result = new String[names.length];

        for (int i = 0; i < names.length; i++) {
            result[i] = names[index[i]];
        }

        return result;
    }
}
```

OUTPUT :

Accepted Runtime: 1 ms

Case 1 Case 2

Input

names =
["Mary","John","Emma"]

heights =
[180,165,170]

Output

["Mary","Emma","John"]

Expected

["Mary","Emma","John"]

Accepted Runtime: 1 ms

Case 1 Case 2

Input

names =
["Alice","Bob","Bob"]

heights =
[155,185,150]

Output

["Bob","Alice","Bob"]

Expected

["Bob","Alice","Bob"]